Dependency-Aware Assessment Generator

Problem Statement

Create a system that generates formative assessments based on the prerequisite concepts of a target learning objective to ensure readiness before progression.

In this web application, our goal is to help learners ensure they're fully prepared before diving into a new topic. Many times, learners struggle because they miss out on the foundational concepts required to understand the material. This system fixes that by analyzing prerequisites and generating customized assessments to evaluate the learner's readiness.

Users interact with the platform in three main ways:

1. **Topic Selection**:
   The user can simply choose a topic they want to learn from a dropdown menu.

2. **Video Upload (YouTube Link)**:
   If the user prefers a specific video lecture, they can paste a YouTube link. The system fetches the transcript of the video and identifies the key concepts covered in it.

3. **PDF Upload**:
   If the user has notes or study material in PDF form, the app extracts the text content and identifies the concepts and their dependencies.

Concept Graph & Prerequisite Mapping

The backbone of this system is a **concept graph**—a structured map where each node is a topic and the edges represent prerequisite relationships. For example, before understanding "Dynamic Programming," the learner should be comfortable with "Recursion" and "Arrays."

AI-Generated Assessments

Once the prerequisites for the selected topic (or video/PDF) are identified, the app uses **Gemini API** to generate quiz questions—typically **5 questions per prerequisite**. These questions are multiple-choice and include correct answers for scoring.

Readiness Evaluation

After taking the quiz:

- **If the learner scores 3 or more out of 5 on each prerequisite**, they're marked as ready and can proceed with confidence.
- **If they score below 3** on any topic, the app:
    - Highlights those **weaker prerequisites** for review.
    - Optionally suggests **alternative topics** they can learn now based on their current knowledge and scores.

TECH STACK

Layer	Tool
Frontend	React.js + Tailwind CSS
Backend	Node.js + Express (or Python FastAPI)
Database	MongoDB (NoSQL, good for storing graph/JSON-like data)
AI API	Gemini API (Google Generative AI for quiz generation)
PDF Parsing	`pdf-parse` (Node.js) or `PyMuPDF` (Python)
YouTube Transcript	`youtube-transcript-api` (Python) or YouTube API
Deployment	Vercel (Frontend), Render or Railway (Backend/API)

✅ System Description – Dependency-Aware Assessment Generator

This web-based learning platform helps learners assess their readiness to study a specific topic by testing their understanding of prerequisite concepts. The system intelligently guides learners by recommending personalized learning paths and dynamically generated assessments.

🔐 User Authentication

- Users begin by signing up or logging into the platform. Each user has a personal profile storing their mastered concepts and assessment history.

🔍 Topic Input Options

Once logged in, the user lands on a dashboard with 3 options to initiate their learning:

1. **Select from a searchable dropdown** of topics (e.g., DSA topics like "Binary Search").
2. **Paste a YouTube video link** — the system extracts its transcript to detect main concepts.
3. **Upload a PDF** — the system analyzes the content textually to find covered topics.

🧠 Concept Extraction & Prerequisite Mapping

- The chosen topic or extracted concept is matched against a **concept graph** (stored in MongoDB) to find **prerequisite topics**.
- These prerequisites are displayed to the user, along with an option to **"Take Prerequisite Quiz."**

📝 AI-Generated Quiz

- A quiz (5/10/15/20 questions per prerequisite) is generated using an **AI API**.
- Users answer the questions within a time limit and click **Submit**.

📊 Score Analysis & Feedback

- Each topic has a threshold (e.g., ≥3/5 or 10/15 or 12/15correct). Based on quiz performance:
  - ✅ **Above threshold**: That topic is marked as **“Mastered”** in the user's profile.
  - ❌ **Below threshold**:
    - AI generates a **capsule summary** of that weak concept for revision.
    - The user can **retake** the quiz for that specific topic.
    - The system also **suggests new topics** that are within the scope of mastered prerequisites.

🎯 Ready for Learning Target Topic

- Once all prerequisites are mastered, a message appears stating the user is "**Ready to Learn the Topic**."

👤 User Profile

- Tracks and displays:
  - ✅ List of mastered topics
  - 🔁 “Start from Scratch” to reset all progress
  - 🌐 Interactive **concept graph** view to see topic relationships

---